CROIP API DOCUMENTATION — FULL ENDPOINT REFERENCE
==========
Total endpoints: 89

POST {zitadel}/oauth/v2/token
Service: ZITADEL
Who/scope: public (client auth)
Description: Concession backend exchanges its client credentials for a short-lived JWT. Sent over mutual TLS.
Request body:
"grant_type=client_credentials&client_id=CON-REVTECH-01&client_secret=***&scope=ref:read property:read sync:write bill:create"
Response body:

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIs...",
  "token_type": "Bearer",
  "expires_in": 900,
  "scope": "ref:read property:read sync:write bill:create"
}
```

Notes: Cache until about 60 seconds before expiry. Revoked or expired api_clients are refused.

GET /api/v1/reference-data
Service: reference
Who/scope: ref:read
Description: Assemblies the caller is contracted to, plus lookup codes. Concessions cache this for offline use.
Request body:
(none)
Response body:

```
{
  "version": "2026-10-15.1",
  "assemblies": [
    {
      "code": "GEMA",
      "name": "Ga East Municipal Assembly",
      "contractStart": "2026-01-01",
      "contractEnd": "2028-12-31"
    }
  ],
  "lookups": {
    "propertyUse": [
```

```
      "RESIDENTIAL",
      "COMMERCIAL"
    ],
    "tenureType": [
      "FREEHOLD",
      "LEASEHOLD"
    ],
    "buildingStatus": [
      "COMPLETED",
      "UNDER_CONSTRUCTION"
    ]
  }
}
Notes: Scope comes from the token, never from a query parameter.

---------------------------------------------------------------------
GET /api/v1/countries
Service: reference
Who/scope: ref:read
Description: List countries.
Request body:
(none)
Response body:
{
  "items": [
    {
      "iso2": "GH",
      "iso3": "GHA",
      "name": "Ghana"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

---------------------------------------------------------------------
POST /api/v1/countries
Service: reference
Who/scope: reference:admin (core admin)
Description: Add a country.
Request body:
{
  "name": "Ghana",
  "iso2": "GH",
  "iso3": "GHA"
```

```
}
Response body:
{
  "iso2": "GH"
}

---------------------------------------------------------------------
GET /api/v1/regions?countryCode=GH
Service: reference
Who/scope: ref:read
Description: List regions. Boundary polygons only with ?include=polygon.
Request body:
(none)
Response body:
{
  "items": [
    {
      "code": "GAR",
      "name": "Greater Accra Region",
      "countryCode": "GH"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

---------------------------------------------------------------------
POST /api/v1/regions
Service: reference
Who/scope: reference:admin
Description: Add a region with its boundary.
Request body:
{
  "countryCode": "GH",
  "name": "Greater Accra Region",
  "code": "GAR",
  "polygon": "GeoJSON MultiPolygon"
}
Response body:
{
  "code": "GAR"
}
Notes: 422 if the boundary is not inside the country boundary where one is held.

---------------------------------------------------------------------
```

```
GET /api/v1/assemblies?regionCode=GAR&type=MUNICIPAL
Service: reference
Who/scope: ref:read
Description: Browse all assemblies.
Request body:
(none)
Response body:
{
  "items": [
    {
      "code": "GEMA",
      "name": "Ga East Municipal Assembly",
      "type": "MUNICIPAL",
      "regionCode": "GAR"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

---------------------------------------------------------------------
GET /api/v1/assemblies/{code}
Service: reference
Who/scope: ref:read
Description: One assembly, optionally with its polygon.
Request body:
(none)
Response body:
{
  "code": "GEMA",
  "name": "Ga East Municipal Assembly",
  "type": "MUNICIPAL",
  "regionCode": "GAR",
  "activeConcessionCode": "REVTECH"
}

---------------------------------------------------------------------
POST /api/v1/assemblies
Service: reference
Who/scope: reference:admin
Description: Add an assembly with its boundary.
Request body:
{
  "regionCode": "GAR",
  "assemblyType": "MUNICIPAL",
```

```
  "name": "Ga East Municipal Assembly",
  "code": "GEMA",
  "polygon": "GeoJSON MultiPolygon"
}
Response body:
{
  "code": "GEMA"
}
Notes: The boundary must sit inside the region. Property and ratepayer assignment
depends on it.

---------------------------------------------------------------------------
PATCH /api/v1/assemblies/{code}
Service: reference
Who/scope: reference:admin
Description: Rename or correct a boundary.
Request body:
{
  "name": "Ga East Municipal Assembly",
  "polygon": "GeoJSON MultiPolygon"
}
Response body:
{
  "code": "GEMA",
  "updated": [
    "polygon"
  ]
}
Notes: Boundary changes publish an event; existing properties are flagged for re-
check, not moved silently.

---------------------------------------------------------------------------
GET /api/v1/lookups/{type}
Service: reference
Who/scope: ref:read
Description: Read a code list. Types: assembly-types, building-statuses, property-
statuses, property-uses, tenure-types, structure-types.
Request body:
(none)
Response body:
{
  "type": "property-uses",
  "items": [
    {
      "code": "RESIDENTIAL",
      "name": "Residential"
```

```
    }
  ]
}

---------------------------------------------------------------------
POST /api/v1/lookups/{type}
Service: reference
Who/scope: reference:admin
Description: Add a code to a list.
Request body:
{
  "code": "MIXED",
  "name": "Mixed use"
}
Response body:
{
  "code": "MIXED"
}

---------------------------------------------------------------------
GET /api/v1/geo/resolve?lat=5.64&lng=-0.18
Service: reference
Who/scope: ref:read
Description: Which assembly and region contain a point. Apps use this before
registering a ratepayer or property.
Request body:
(none)
Response body:
{
  "assemblyCode": "GEMA",
  "regionCode": "GAR",
  "inConcessionScope": true
}

---------------------------------------------------------------------
POST /api/v1/concessions
Service: concession
Who/scope: concession:admin (core admin)
Description: Onboard a concession.
Request body:
{
  "code": "REVTECH",
  "name": "RevTech Ltd",
  "contactEmail": "ops@revtech.example"
}
Response body:
```

```
{
  "code": "REVTECH",
  "status": "ACTIVE"
}
```

POST /api/v1/concessions/{code}/contracts
Service: concession
Who/scope: concession:admin
Description: Assign an assembly to a concession for a date range.
Request body:

```
{
  "assemblyCode": "GEMA",
  "contractStart": "2026-01-01",
  "contractEnd": "2028-12-31"
}
```

Response body:

```
{
  "contractId": "ctr_01",
  "status": "ACTIVE"
}
```

Notes: 409 CONTRACT_OVERLAP if another concession holds that assembly in an overlapping period.

POST /api/v1/concessions/{code}/personnel
Service: concession
Who/scope: concession:admin or concession self-service
Description: Register a field staff member so every verification traces to a named person.
Request body:

```
{
  "externalStaffRef": "STAFF-0042",
  "fullName": "Ama Mensah",
  "phone": "+233240000000"
}
```

Response body:

```
{
  "staffRef": "STAFF-0042",
  "status": "ACTIVE"
}
```

Notes: Personnel have no CROIP login. Set status REVOKED with PATCH on the same resource to remove access.

POST /api/v1/concessions/{code}/api-clients

Service: concession
Who/scope: concession:admin
Description: Issue API credentials. The secret is shown once and only its hash is stored.
Request body:

```
{
  "expiresAt": "2027-10-15T00:00:00Z"
}
```

Response body:

```
{
  "clientId": "CON-REVTECH-01",
  "clientSecret": "shown-once",
  "status": "ACTIVE",
  "expiresAt": "2027-10-15T00:00:00Z"
}
```

Notes: Revoke with POST /api/v1/api-clients/{clientId}/revoke. Revoked clients fail at the next token request.

GET /api/v1/concessions
Service: concession
Who/scope: concession:admin or oversight
Description: List concessions.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "code": "REVTECH",
      "name": "RevTech Ltd",
      "status": "ACTIVE"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

GET /api/v1/concessions/{code}
Service: concession
Who/scope: concession:read
Description: One concession with contracts and counts. A concession reads only itself.
Request body:

(none)
Response body:

```
{
  "code": "REVTECH",
  "status": "ACTIVE",
  "contracts": 1,
  "personnel": 12,
  "apiClients": 1
}
```

PATCH /api/v1/concessions/{code}
Service: concession
Who/scope: concession:admin
Description: Change contact details or status.
Request body:

```
{
  "status": "SUSPENDED",
  "reason": "Contract review"
}
```

Response body:

```
{
  "code": "REVTECH",
  "status": "SUSPENDED"
}
```

Notes: Suspended or terminated concessions fail at the next token request and their active payment links stay valid until they expire.

GET /api/v1/concessions/{code}/contracts
Service: concession
Who/scope: concession:read
Description: Contract history per assembly.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "contractId": "ctr_01",
      "assemblyCode": "GEMA",
      "contractStart": "2026-01-01",
      "contractEnd": "2028-12-31",
      "status": "ACTIVE"
    }
  ],
```

```
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

PATCH /api/v1/concessions/{code}/contracts/{contractId}
Service: concession
Who/scope: concession:admin
Description: Terminate or change the end date.
Request body:

```
{
  "status": "TERMINATED",
  "contractEnd": "2027-06-30"
}
```

Response body:

```
{
  "contractId": "ctr_01",
  "status": "TERMINATED"
}
```

Notes: Still checked against overlap. Later contracts for the same assembly start after this end date.

GET /api/v1/concessions/{code}/personnel
Service: concession
Who/scope: concession:read
Description: List field staff.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "staffRef": "STAFF-0042",
      "fullName": "Ama Mensah",
      "status": "ACTIVE"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

PATCH /api/v1/concessions/{code}/personnel/{staffRef}

Service: concession
Who/scope: concession:admin or concession self-service
Description: Update details or set INACTIVE or REVOKED.
Request body:

```
{
  "status": "REVOKED"
}
```

Response body:

```
{
  "staffRef": "STAFF-0042",
  "status": "REVOKED"
}
```

Notes: Past verifications and batches keep pointing to the staff record.

GET /api/v1/concessions/{code}/api-clients
Service: concession
Who/scope: concession:read
Description: List credentials. Secrets are never returned.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "clientId": "CON-REVTECH-01",
      "status": "ACTIVE",
      "expiresAt": "2027-10-15T00:00:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

POST /api/v1/api-clients/{clientId}/revoke
Service: concession
Who/scope: concession:admin
Description: Revoke credentials now.
Request body:

```
{
  "reason": "Key exposed"
}
```

Response body:

```
{
```

```
  "clientId": "CON-REVTECH-01",
  "status": "REVOKED"
}

---------------------------------------------------------------------
POST /api/v1/users
Service: identity-admin
Who/scope: user:admin (core admin)
Description: Onboard a government oversight user for a country. Creates the
ZITADEL account and the CROIP record.
Request body:
{
  "email": "officer@gov.example",
  "fullName": "Efua Boateng",
  "countryCode": "GH",
  "phone": "+233500000000"
}
Response body:
{
  "code": "USR-0007",
  "status": "ACTIVE"
}
Notes: Roles are assigned in ZITADEL. Concession staff and ratepayers are never
created here.

---------------------------------------------------------------------
GET /api/v1/users?countryCode=GH
Service: identity-admin
Who/scope: user:admin
Description: List oversight users.
Request body:
(none)
Response body:
{
  "items": [
    {
      "code": "USR-0007",
      "fullName": "Efua Boateng",
      "countryCode": "GH",
      "status": "ACTIVE"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

GET /api/v1/users/{code}
Service: identity-admin
Who/scope: user:admin or self
Description: One user.
Request body:
(none)
Response body:

```
{
  "code": "USR-0007",
  "email": "officer@gov.example",
  "countryCode": "GH",
  "status": "ACTIVE"
}
```

PATCH /api/v1/users/{code}
Service: identity-admin
Who/scope: user:admin
Description: Change details, or set SUSPENDED or DEACTIVATED (also applied in ZITADEL).
Request body:

```
{
  "status": "SUSPENDED"
}
```

Response body:

```
{
  "code": "USR-0007",
  "status": "SUSPENDED"
}
```

POST /api/v1/ratepayers/register
Service: ratepayer
Who/scope: ratepayer:register (signed-in new account)
Description: Ratepayer registers themselves. They sign up in ZITADEL first, then call this with that token. Region and assembly are found from the map pin.
Request body:

```
{
  "ratepayerType": "INDIVIDUAL",
  "legalName": "Kofi Darko",
  "phone": "+233200000000",
  "email": "kofi@example.com",
  "location": {
    "type": "Point",
```

```
    "coordinates": [
      -0.18,
      5.64
    ]
  }
}
Response body:
{
  "code": "RP-2026-00001",
  "assemblyCode": "GEMA",
  "regionCode": "GAR",
  "idVerificationStatus": "NONE"
}
Notes: 422 LOCATION_OUTSIDE_ASSEMBLIES if the pin is not inside any assembly. The
token subject is linked to the new ratepayer.

---------------------------------------------------------------------
POST /api/v1/ratepayers
Service: ratepayer
Who/scope: ratepayer:create (concession)
Description: Concession onboards a ratepayer inside its contracted assemblies. The
ratepayer can claim a login later.
Request body:
{
  "ratepayerType": "INDIVIDUAL",
  "legalName": "Kofi Darko",
  "phone": "+233200000000",
  "email": "kofi@example.com",
  "location": {
    "type": "Point",
    "coordinates": [
      -0.18,
      5.64
    ]
  },
  "clientEventId": "5a1c2e7e-3a43-4a6e-b1a0-1f0c7d1b2201"
}
Response body:
{
  "code": "RP-2026-00002",
  "assemblyCode": "GEMA",
  "claimInviteSent": true,
  "possibleDuplicates": [
    {
      "code": "RP-2026-00001",
      "reason": "SAME_PHONE"
```

```
    }
  ]
}
Notes: 403 if the pin is outside the concession scope. A phone number is a
contact, not an identity, so matches are returned for review, not rejected.
Idempotent on clientEventId.

---------------------------------------------------------------------------
GET /api/v1/ratepayers?
assemblyCode=GEMA&idVerified=false&hasProperty=false&q=darko&page=1
Service: ratepayer
Who/scope: ratepayer:read
Description: Concession sees every ratepayer in its scope, even those with no
property yet. Oversight users see their country.
Request body:
(none)
Response body:
{
  "items": [
    {
      "code": "RP-2026-00001",
      "legalName": "Kofi Darko",
      "type": "INDIVIDUAL",
      "phone": "+233200000000",
      "assemblyCode": "GEMA",
      "idVerificationStatus": "VERIFIED",
      "propertyCount": 1
    }
  ],
  "page": 1,
  "nextPage": 2,
  "total": 5230
}
Notes: Filters are optional. National ID numbers are never returned. A ratepayer
token returns only that ratepayer.

---------------------------------------------------------------------------
GET /api/v1/ratepayers/{code}
Service: ratepayer
Who/scope: ratepayer:read
Description: One ratepayer with linked properties and bills summary.
Request body:
(none)
Response body:
{
  "code": "RP-2026-00001",
```

```
  "legalName": "Kofi Darko",
  "assemblyCode": "GEMA",
  "properties": [
    {
      "code": "GA-014-8821",
      "ownershipType": "OWNER",
      "percentage": "100.00"
    }
  ]
}
```

PATCH /api/v1/ratepayers/{code}
Service: ratepayer
Who/scope: ratepayer:update
Description: Update contact details or location. Changing the pin re-resolves the assembly.
Request body:

```
{
  "phone": "+233200000001",
  "location": {
    "type": "Point",
    "coordinates": [
      -0.19,
      5.65
    ]
  }
}
```

Response body:

```
{
  "code": "RP-2026-00001",
  "assemblyCode": "GEMA",
  "updated": [
    "phone",
    "location"
  ]
}
```

Notes: A concession can only update ratepayers inside its scope, and only until the pin moves them out.

POST /api/v1/ratepayers/{code}/id-verifications
Service: ratepayer
Who/scope: ratepayer:update
Description: Start third-party ID verification (for example Ghana Card through an ID provider). Only the provider and its reference are stored.

Request body:

```
{
  "provider": "GHANA_CARD_NIA",
  "providerReference": "tok_9f31a"
}
```

Response body:

```
{
  "id": "idv_01",
  "verificationStatus": "PENDING"
}
```

Notes: The provider result arrives by callback and updates the status to VERIFIED or FAILED. A ratepayer needs one VERIFIED check before managing assessed properties.

POST /api/v1/properties/{code}/claims

Service: ratepayer

Who/scope: ratepayer:create (self)

Description: Claim an unclaimed property — one with no ratepayer link yet, usually created directly by a concession. States the ownership relationship and attaches evidence documents, so the concession can review the claim from the documents rather than visiting.

Request body:

```
{
  "relationship": "OWNER",
  "ownershipSharePercent": 100,
  "evidenceDocuments": [
    {
      "docType": "DEED",
      "mediaUrl": "https://media.example/docs/deed-9003.pdf"
    }
  ]
}
```

Response body:

```
{
  "linkId": "lnk_14",
  "propertyCode": "GA-014-9003",
  "ratepayerCode": "RP-2026-00118",
  "relationship": "OWNER",
  "claimStatus": "PENDING_REVIEW"
}
```

Notes: Only works if the property currently has zero ratepayer links. The concession reviews the documents and verifies through the normal property-verification step; it only needs to visit if the documents don't settle it.

GET /api/v1/me/bills/{billNumber}
Service: ratepayer
Who/scope: ratepayer:read (self)
Description: Look up a bill by its code to pay on someone else's behalf. Returns just enough to confirm it's the right bill, not the other ratepayer's full profile.
Request body:
(none)
Response body:

```
{
  "billNumber": "BILL-2026-GA-0891",
  "propertyCode": "GA-014-8821",
  "ratepayerName": "Kojo Mensah",
  "amountDue": "1200.00",
  "amountPaid": "300.00",
  "balance": "900.00"
}
```

Notes: Any authenticated ratepayer can look up any bill by its exact code; the code itself is the permission, like paying someone's electricity bill with their meter number.

POST /api/v1/bills/{billNumber}/payment-links
Service: ratepayer
Who/scope: ratepayer:create (self or paying for another)
Description: Reserve an exact amount against a bill, full or partial, paid by the ratepayer on the bill or by someone paying on their behalf.
Request body:

```
{
  "amount": "900.00",
  "paidByRatepayerCode": "RP-2026-00118"
}
```

Response body:

```
{
  "linkId": "lnk_09",
  "billNumber": "BILL-2026-GA-0891",
  "amount": "900.00",
  "paidByRatepayerCode": "RP-2026-00118",
  "status": "ACTIVE",
  "expiresAt": "2026-10-15T23:00:00Z"
}
```

Notes: paidByRatepayerCode defaults to the caller. The lock still applies against the bill's own remaining balance, so two people paying the same bill still cannot jointly overpay it.

GET /api/v1/me/properties?page=1
Service: ratepayer
Who/scope: ratepayer:read (self)
Description: Every property the ratepayer owns, co-owns, rents or has any link to, across every concession, each tagged with the concession that manages it.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "propertyCode": "GA-014-8821",
      "role": "OWNER",
      "assemblyCode": "GEMA",
      "concessionCode": "REVTECH",
      "concessionName": "RevTech Ltd",
      "verificationStatus": "VERIFIED"
    },
    {
      "propertyCode": "KS-002-0417",
      "role": "CO_OWNER",
      "assemblyCode": "KMA",
      "concessionCode": "UrbanRate",
      "concessionName": "UrbanRate Services",
      "verificationStatus": "PENDING"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 2
}
```

Notes: One ratepayer can have properties under several concessions; this is the one view that crosses concession scope, because it belongs to the ratepayer, not to a concession.

GET /api/v1/ratepayers/{code}/id-verifications
Service: ratepayer
Who/scope: ratepayer:read
Description: ID verification history for one ratepayer.
Request body:
(none)
Response body:

```
{
  "items": [
    {
```

```
      "id": "idv_01",
      "provider": "GHANA_CARD_NIA",
      "verificationStatus": "VERIFIED",
      "verifiedAt": "2026-10-15T08:00:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

```
POST /api/v1/id-verifications/callback
Service: ratepayer
Who/scope: provider credential
Description: The ID provider reports a result.
Request body:
{
  "providerReference": "tok_9f31a",
  "status": "VERIFIED",
  "verifiedAt": "2026-10-15T08:00:00Z"
}
Response body:
{
  "id": "idv_01",
  "verificationStatus": "VERIFIED"
}
Notes: Signed by the provider. Never carries ID numbers or biometrics.
```

```
POST /api/v1/properties
Service: property-registry
Who/scope: property:create (ratepayer)
Description: Ratepayer registers a property, stating their ownership relationship
and attaching evidence documents (deed, tenancy agreement, utility bill). It
starts PENDING and UNASSESSED. The concession can verify remotely from the
documents, without a site visit, unless something looks off.
Request body:
{
  "propertyName": "Darko Villa",
  "propertyUse": "RESIDENTIAL",
  "tenureType": "FREEHOLD",
  "structureType": "CONCRETE",
  "buildingStatus": "COMPLETED",
  "numberOfFloors": 2,
  "landAreaSqm": 450.5,
```

```
"polygon": {
  "type": "MultiPolygon",
  "coordinates": [
    [
      [
        [
          -0.1801,
          5.6401
        ],
        [
          -0.1799,
          5.6401
        ],
        [
          -0.1799,
          5.6399
        ],
        [
          -0.1801,
          5.6399
        ],
        [
          -0.1801,
          5.6401
        ]
      ]
    ]
  ]
},
"units": [
  {
    "unitIdentifier": "MAIN",
    "unitType": "WHOLE_BUILDING"
  }
],
"ownership": {
  "relationship": "OWNER",
  "ownershipSharePercent": 100,
  "evidenceDocuments": [
    {
      "docType": "DEED",
      "mediaUrl": "https://media.example/docs/deed-8821.pdf"
    },
    {
      "docType": "UTILITY_BILL",
      "mediaUrl": "https://media.example/docs/ecg-8821.pdf"
```

```
      }
    ]
  }
}
Response body:
{
  "code": "GA-014-8821",
  "accountNumber": null,
  "verificationStatus": "PENDING",
  "propertyStatus": "UNASSESSED",
  "assemblyCode": "GEMA"
}
Notes: 422 if the polygon is not inside an assembly. Assembly is found with
ST_Covers. Polygon is GeoJSON, stored as geometry(MultiPolygon,4326).

---------------------------------------------------------------------
POST /api/v1/properties
Service: property-registry
Who/scope: property:create (concession)
Description: Concession registers a property for a ratepayer in its scope. Same
path as above; the token decides which rules apply.
Request body:
{
  "propertyName": "Darko Villa",
  "propertyUse": "RESIDENTIAL",
  "tenureType": "FREEHOLD",
  "structureType": "CONCRETE",
  "buildingStatus": "COMPLETED",
  "numberOfFloors": 2,
  "landAreaSqm": 450.5,
  "polygon": {
    "type": "MultiPolygon",
    "coordinates": [
      [
        [
          [
            -0.1801,
            5.6401
          ],
          [
            -0.1799,
            5.6401
          ],
          [
            -0.1799,
            5.6399
```

```
          ],
          [
            -0.1801,
            5.6399
          ],
          [
            -0.1801,
            5.6401
          ]
        ]
      ]
    ]
  },
  "units": [
    {
      "unitIdentifier": "MAIN",
      "unitType": "WHOLE_BUILDING"
    }
  ],
  "ownership": {
    "relationship": "OWNER",
    "ownershipSharePercent": 100,
    "evidenceDocuments": [
      {
        "docType": "DEED",
        "mediaUrl": "https://media.example/docs/deed-8821.pdf"
      },
      {
        "docType": "UTILITY_BILL",
        "mediaUrl": "https://media.example/docs/ecg-8821.pdf"
      }
    ]
  },
  "clientEventId": "3c1f1d62-2d9b-4a5e-8e1a-5a6e0a7b3302",
  "ratepayerCode": "RP-2026-00002",
  "ownershipType": "OWNER"
}
Response body:
{
  "code": "GA-014-8821",
  "accountNumber": null,
  "verificationStatus": "PENDING",
  "propertyStatus": "UNASSESSED",
  "assemblyCode": "GEMA"
}
Notes: 403 if the polygon is outside the concession scope. Idempotent on
```

clientEventId. Verification is still a separate step.

GET /api/v1/properties?
assemblyCode=GEMA&verificationStatus=PENDING&propertyStatus=UNASSESSED&ratepayerCoc
2026-00002&page=1
Service: property-registry
Who/scope: property:read
Description: Concession sees all properties in its scope. Filter by PENDING to get the verification work queue.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "code": "GA-014-8821",
      "propertyName": "Darko Villa",
      "assemblyCode": "GEMA",
      "verificationStatus": "PENDING",
      "propertyStatus": "UNASSESSED",
      "centroid": {
        "type": "Point",
        "coordinates": [
          -0.18,
          5.64
        ]
      }
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

Notes: Ratepayers see only properties linked to them.

GET /api/v1/properties/{code}
Service: property-registry
Who/scope: property:read
Description: One property with units, owners and current valuation summary. Outside scope returns 404.
Request body:
(none)
Response body:

```
{
```

```
  "code": "GA-014-8821",
  "accountNumber": "GEMA 01 024 008",
  "verificationStatus": "VERIFIED",
  "propertyStatus": "ASSESSED",
  "units": [
    {
      "unitIdentifier": "MAIN",
      "unitType": "WHOLE_BUILDING"
    }
  ],
  "owners": [
    {
      "ratepayerCode": "RP-2026-00001",
      "ownershipType": "OWNER",
      "percentage": "100.00"
    }
  ],
  "currentValuation": {
    "rateableValue": "48000.00",
    "lvdReference": "LVD-2026-1182"
  }
}
```

PATCH /api/v1/properties/{code}
Service: property-registry
Who/scope: property:update
Description: Change property details.
Request body:

```
{
  "propertyUse": "COMMERCIAL",
  "numberOfFloors": 3
}
```

Response body:

```
{
  "code": "GA-014-8821",
  "verificationStatus": "PENDING",
  "updated": [
    "propertyUse",
    "numberOfFloors"
  ]
}
```

Notes: Changing a verified attribute resets verification to PENDING, so it must be verified again before the next bill.

POST /api/v1/properties/{code}/ratepayer-links
Service: property-registry
Who/scope: property:link
Description: Link a ratepayer to a unit as owner, co-owner, tenant, caretaker or agent.
Request body:

```
{
  "unitIdentifier": "MAIN",
  "ratepayerCode": "RP-2026-00002",
  "ownershipType": "CO_OWNER",
  "ownershipPercentage": "50.00",
  "isPrimaryBillingContact": false,
  "effectiveFrom": "2026-10-15"
}
```

Response body:

```
{
  "linkId": "lnk_01",
  "unitIdentifier": "MAIN",
  "ownershipPercentage": "50.00"
}
```

Notes: 409 if shares on the unit would exceed 100 or a second primary billing contact is set.

POST /api/v1/property-verifications
Service: property-registry
Who/scope: property:verify
Description: Concession surveyor confirms what the ratepayer submitted. Idempotent on clientEventId.
Request body:

```
{
  "clientEventId": "0d9c5e0e-6c5b-4d0a-9a53-0d5f0f3b7a11",
  "propertyCode": "GA-014-8821",
  "staffRef": "STAFF-0042",
  "verifierLocation": {
    "type": "Point",
    "coordinates": [
      -0.18,
      5.64
    ]
  },
  "capturedAt": "2026-10-15T09:12:00Z",
  "outcome": "ACCEPTED",
  "verifiedSnapshot": {
    "numberOfFloors": 2,
    "propertyUse": "RESIDENTIAL"
```

```
  },
  "accountNumber": "GEMA 01 024 008"
}
Response body:
{
  "id": "ver_01",
  "isWithinPolygon": true,
  "outcome": "ACCEPTED",
  "propertyVerificationStatus": "VERIFIED"
}
Notes: Server computes isWithinPolygon. Out-of-polygon results are flagged for
audit. VERIFIED is applied via PropertyVerified event.

---------------------------------------------------------------------
POST /api/v1/properties/{code}/photos
Service: property-registry
Who/scope: property:update
Description: Attach geotagged evidence photo metadata. Media storage is not
designed yet, so this takes a URL.
Request body:
{
  "photoType": "FRONTAGE",
  "mediaUrl": "https://media.example/p/123.jpg",
  "capturedAt": "2026-10-15T09:10:00Z"
}
Response body:
{
  "photoId": "pho_01"
}

---------------------------------------------------------------------
POST /api/v1/properties/{code}/units
Service: property-registry
Who/scope: property:update
Description: Add a unit (apartment, shop, office).
Request body:
{
  "unitIdentifier": "SHOP-1",
  "unitType": "SHOP",
  "floorLevel": 0
}
Response body:
{
  "unitIdentifier": "SHOP-1"
}
Notes: Unit identifier is unique within the property.
```

PATCH /api/v1/properties/{code}/ratepayer-links/{linkId}
Service: property-registry
Who/scope: property:link
Description: End a link (sale, tenancy end) or change the share. History is kept.
Request body:

```
{
  "effectiveTo": "2026-12-31"
}
```

Response body:

```
{
  "linkId": "lnk_01",
  "effectiveTo": "2026-12-31"
}
```

Notes: Past liability and payments stay attached to the ratepayer who held the link.

GET /api/v1/properties/{code}/photos
Service: property-registry
Who/scope: property:read
Description: Evidence photos for one property.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "photoId": "pho_01",
      "photoType": "FRONTAGE",
      "mediaUrl": "https://media.example/p/123.jpg",
      "capturedAt": "2026-10-15T09:10:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

GET /api/v1/property-verifications?propertyCode=GA-014-8821&outcome=ACCEPTED&isWithinPolygon=false
Service: property-registry
Who/scope: property:read
Description: Verification history, for audit and out-of-polygon follow-up.

Request body:
(none)
Response body:

```
{
  "items": [
    {
      "id": "ver_01",
      "propertyCode": "GA-014-8821",
      "staffRef": "STAFF-0042",
      "isWithinPolygon": true,
      "outcome": "ACCEPTED",
      "capturedAt": "2026-10-15T09:12:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

GET /api/v1/property-verifications/{id}
Service: property-registry
Who/scope: property:read
Description: One verification with its snapshot.
Request body:
(none)
Response body:

```
{
  "id": "ver_01",
  "outcome": "ACCEPTED",
  "verifiedSnapshot": {
    "numberOfFloors": 2
  },
  "receivedAt": "2026-10-15T09:13:00Z"
}
```

POST /api/v1/valuation-lists
Service: valuation
Who/scope: valuation:publish (LVD or oversight)
Description: Open a valuation list or supplementary list.
Request body:

```
{
  "name": "GEMA Supplementary 2026",
  "effectiveDate": "2026-11-01"
}
```

Response body:

```
{
  "name": "GEMA Supplementary 2026",
  "status": "DRAFT"
}
```

POST /api/v1/valuation-lists/{name}/valuations
Service: valuation
Who/scope: valuation:publish
Description: Load rateable values. Each item has its own result. Accepted values turn properties ASSESSED.
Request body:

```
{
  "items": [
    {
      "propertyCode": "GA-014-8821",
      "rateableValue": "48000.00",
      "lvdReference": "LVD-2026-1182",
      "effectiveDate": "2026-11-01"
    }
  ]
}
```

Response body:

```
{
  "accepted": 1,
  "results": [
    {
      "propertyCode": "GA-014-8821",
      "status": "APPLIED"
    }
  ]
}
```

Notes: Valuations are never overwritten; a new row becomes current.

GET /api/v1/valuation-lists
Service: valuation
Who/scope: valuation:read
Description: List valuation lists.
Request body:
(none)
Response body:

```
{
  "items": [
    {
```

```
      "name": "GEMA Supplementary 2026",
      "effectiveDate": "2026-11-01",
      "status": "ACTIVE"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

----------------------------------------------------------------------
PATCH /api/v1/valuation-lists/{name}
Service: valuation
Who/scope: valuation:publish
Description: Activate a draft or supersede a list.
Request body:
{
  "status": "ACTIVE"
}
Response body:
{
  "name": "GEMA Supplementary 2026",
  "status": "ACTIVE"
}

----------------------------------------------------------------------
GET /api/v1/properties/{code}/valuations
Service: valuation
Who/scope: valuation:read
Description: Valuation history for a property. The current row is flagged.
Request body:
(none)
Response body:
{
  "items": [
    {
      "valuationList": "GEMA Supplementary 2026",
      "rateableValue": "48000.00",
      "isCurrent": true,
      "lvdReference": "LVD-2026-1182"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

POST /api/v1/sync-batches
Service: sync
Who/scope: sync:write
Description: Offline upload. Every item has its own result; one bad item never drops the rest.
Request body:

```
{
  "deviceId": "TAB-0098",
  "staffRef": "STAFF-0042",
  "items": [
    {
      "type": "property-verification",
      "clientEventId": "0d9c5e0e-6c5b-4d0a-9a53-0d5f0f3b7a11",
      "capturedAt": "2026-10-15T09:12:00Z",
      "body": {
        "propertyCode": "GA-014-8821",
        "outcome": "ACCEPTED"
      }
    },
    {
      "type": "payment",
      "clientEventId": "7b8f1a2c-1e33-4cb2-8a26-6a3c9d7a0001",
      "capturedAt": "2026-10-15T09:40:00Z",
      "body": {
        "billNumber": "BILL-2026-GA-0891",
        "amount": "300.00"
      }
    }
  ]
}
```

Response body:

```
{
  "batchId": "bat_01",
  "status": "COMPLETED",
  "itemCount": 2,
  "results": [
    {
      "clientEventId": "0d9c...",
      "status": "APPLIED"
    },
    {
      "clientEventId": "7b8f...",
      "status": "QUARANTINED",
      "reason": "UNKNOWN_BILL"
```

```
    }
  ]
}
Notes: HTTP 202 means durable receipt only. Item status says what was applied.
Same clientEventId with a different payload is a conflict.

---------------------------------------------------------------------
GET /api/v1/ingestions/{id}
Service: sync
Who/scope: sync:write
Description: Check an earlier submission: received, validated, applied or
quarantined.
Request body:
(none)
Response body:
{
  "id": "bat_01",
  "status": "APPLIED",
  "quarantined": []
}

---------------------------------------------------------------------
GET /api/v1/sync-batches?status=FAILED
Service: sync
Who/scope: sync:read
Description: Batches received from a concession (own only), or all for oversight.
Request body:
(none)
Response body:
{
  "items": [
    {
      "batchId": "bat_01",
      "deviceId": "TAB-0098",
      "staffRef": "STAFF-0042",
      "status": "COMPLETED",
      "itemCount": 2,
      "receivedAt": "2026-10-15T10:20:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

---------------------------------------------------------------------
```

GET /api/v1/sync-batches/{id}
Service: sync
Who/scope: sync:read
Description: One batch with its per-item results.
Request body:
(none)
Response body:

```
{
  "batchId": "bat_01",
  "status": "COMPLETED",
  "results": [
    {
      "clientEventId": "0d9c...",
      "status": "APPLIED"
    }
  ]
}
```

POST /api/v1/bills
Service: billing
Who/scope: bill:create
Description: Concession issues a bill. Only VERIFIED properties are billable; an unassessed bill is allowed before LVD data arrives.
Request body:

```
{
  "externalBillRef": "REV-889102",
  "propertyCode": "GA-014-8821",
  "ratepayerCode": "RP-2026-00001",
  "billingPeriod": "2026",
  "billType": "UNASSESSED",
  "currency": "GHS",
  "amountDue": "1200.00",
  "dueDate": "2026-12-31",
  "valuationRef": null
}
```

Response body:

```
{
  "billNumber": "BILL-2026-GA-0891",
  "paymentStatus": "UNPAID",
  "amountDue": "1200.00",
  "amountPaid": "0.00",
  "supersedes": null
}
```

Notes: One live bill per property per period. 409 on duplicates. An assessed bill later replaces a provisional one and carries payments over.

GET /api/v1/bills/{billNumber}
Service: billing
Who/scope: bill:read
Description: Bill with balance and reserved amount.
Request body:
(none)
Response body:

```
{
  "billNumber": "BILL-2026-GA-0891",
  "amountDue": "1200.00",
  "amountPaid": "300.00",
  "reservedByActiveLinks": "600.00",
  "available": "300.00",
  "paymentStatus": "PARTIALLY_PAID"
}
```

POST /api/v1/bills/{billNumber}/payment-links
Service: billing
Who/scope: paylink:create
Description: Reserves an exact amount and returns the reference the concession must put on its gateway checkout. The ratepayer never types an amount.
Request body:

```
{
  "amount": "600.00"
}
```

Response body:

```
{
  "linkId": "lnk_01",
  "amountFixed": "600.00",
  "status": "ACTIVE",
  "expiresAt": "2026-10-15T10:15:00Z",
  "gatewayLinkRef": "CRL-7F3A91"
}
```

Notes: Bill row is locked. Rejected with 409 BALANCE_EXCEEDED unless amount is at most due minus paid minus active links. Expires after 15 to 30 minutes and frees the balance.

POST /api/v1/payments
Service: billing
Who/scope: payment:report
Description: Concession reports a gateway outcome. Money has already moved, so a confirmed payment is never rejected. Header Idempotency-Key required.

Request body:

```
{
  "clientEventId": "7b8f1a2c-1e33-4cb2-8a26-6a3c9d7a0001",
  "gatewayLinkRef": "CRL-7F3A91",
  "externalPaymentRef": "MOMO-739221",
  "paymentType": "PAYMENT",
  "amount": "600.00",
  "status": "CONFIRMED",
  "paidAt": "2026-10-15T10:02:00Z"
}
```

Response body:

```
{
  "paymentId": "pay_01",
  "billNumber": "BILL-2026-GA-0891",
  "balanceAtPayment": "300.00",
  "paymentStatus": "PARTIALLY_PAID",
  "linkStatus": "PAID"
}
```

Notes: Use paymentType REFUND, CREDIT_IN or CREDIT_OUT with relatedPaymentId for overpayment resolution. Any overpayment opens a reconciliation item.

GET /api/v1/bills?assemblyCode=GEMA&paymentStatus=UNPAID&billingPeriod=2026&propertyCode=&ratepayerCc

Service: billing

Who/scope: bill:read

Description: All bills in scope. Ratepayers see only their own.

Request body:

(none)

Response body:

```
{
  "items": [
    {
      "billNumber": "BILL-2026-GA-0891",
      "propertyCode": "GA-014-8821",
      "amountDue": "1200.00",
      "amountPaid": "0.00",
      "paymentStatus": "UNPAID",
      "dueDate": "2026-12-31"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

POST /api/v1/bills/{billNumber}/supersede
Service: billing
Who/scope: bill:create
Description: Replace a bill, for example an assessed bill replacing an unassessed or provisional one after LVD values arrive.
Request body:

```
{
  "externalBillRef": "REV-889140",
  "billType": "ASSESSED",
  "valuationRef": "LVD-2026-1182",
  "amountDue": "1450.00"
}
```

Response body:

```
{
  "billNumber": "BILL-2026-GA-0902",
  "supersedes": "BILL-2026-GA-0891",
  "carriedOverCredit": "300.00",
  "cancelledLinks": 1
}
```

Notes: Active payment links on the old bill are cancelled in the same step. Payments carry over as credit.

GET /api/v1/bills/{billNumber}/payment-links
Service: billing
Who/scope: bill:read
Description: Links for a bill, active and past.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "linkId": "lnk_01",
      "amountFixed": "600.00",
      "status": "ACTIVE",
      "expiresAt": "2026-10-15T10:15:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

POST /api/v1/payment-links/{linkId}/cancel
Service: billing
Who/scope: paylink:create
Description: Cancel an active link and free the reserved balance.
Request body:

```
{
  "reason": "Ratepayer asked for a different amount"
}
```

Response body:

```
{
  "linkId": "lnk_01",
  "status": "CANCELLED"
}
```

Notes: Only ACTIVE links can be cancelled. The concession must also void its gateway checkout.

GET /api/v1/payments?billNumber=BILL-2026-GA-0891&status=CONFIRMED&from=2026-10-01
Service: billing
Who/scope: payment:read
Description: Payments in scope, with refunds and credit transfers.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "paymentId": "pay_01",
      "billNumber": "BILL-2026-GA-0891",
      "paymentType": "PAYMENT",
      "amount": "600.00",
      "status": "CONFIRMED",
      "paidAt": "2026-10-15T10:02:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
```

GET /api/v1/payments/{paymentId}
Service: billing
Who/scope: payment:read
Description: One payment with its related refund or credit entries.
Request body:

(none)
Response body:

```
{
  "paymentId": "pay_01",
  "externalPaymentRef": "MOMO-739221",
  "amount": "600.00",
  "status": "CONFIRMED",
  "related": []
}
```

POST /api/v1/payments/{paymentId}/reversals
Service: billing
Who/scope: payment:report
Description: Report that the gateway reversed a payment. Original record is kept; the bill balance is corrected.
Request body:

```
{
  "clientEventId": "c2d8e1f0-7d0a-42b1-9a30-3f4a9d1e0004",
  "externalPaymentRef": "MOMO-739221-R",
  "reason": "Chargeback",
  "reversedAt": "2026-10-16T08:00:00Z"
}
```

Response body:

```
{
  "paymentId": "pay_02",
  "status": "REVERSED",
  "paymentStatus": "PARTIALLY_PAID"
}
```

Notes: Idempotency-Key required.

POST /api/v1/control-totals
Service: reconciliation
Who/scope: control:write
Description: Period manifest for completeness checks: counts and amounts by event family.
Request body:

```
{
  "periodStart": "2026-10-01T00:00:00Z",
  "periodEnd": "2026-10-15T00:00:00Z",
  "currency": "GHS",
  "totals": {
    "billsIssued": {
      "count": 1200,
      "minor": 144000000
```

```
    },
    "paymentsConfirmed": {
      "count": 410,
      "minor": 49200000
    },
    "verifications": {
      "count": 380
    }
  }
}
Response body:
{
  "id": "ctl_01",
  "status": "RECEIVED",
  "differences": []
}

---------------------------------------------------------------------
GET /api/v1/exceptions?status=OPEN
Service: reconciliation
Who/scope: exception:read
Description: Reconciliation discrepancies. Concessions see their own; oversight
users see their country.
Request body:
(none)
Response body:
{
  "items": [
    {
      "id": "exc_01",
      "type": "OVERPAYMENT",
      "billNumber": "BILL-2026-GA-0891",
      "status": "OPEN",
      "resolution": null
    }
  ]
}
Notes: Resolving needs a named user. No write-offs: refund or carry credit
forward.

---------------------------------------------------------------------
POST /api/v1/reconciliation-runs
Service: reconciliation
Who/scope: recon:run (oversight user)
Description: Run bills vs verifications vs payments for one concession and period.
Request body:
```

```
{
  "concessionCode": "REVTECH",
  "periodStart": "2026-10-01T00:00:00Z",
  "periodEnd": "2026-10-15T00:00:00Z"
}
Response body:
{
  "runId": "run_01",
  "status": "RUNNING"
}

---------------------------------------------------------------------------
GET /api/v1/reconciliation-runs?concessionCode=REVTECH
Service: reconciliation
Who/scope: recon:read
Description: Run history.
Request body:
(none)
Response body:
{
  "items": [
    {
      "runId": "run_01",
      "concessionCode": "REVTECH",
      "status": "COMPLETED",
      "openItems": 3
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}

---------------------------------------------------------------------------
GET /api/v1/reconciliation-runs/{runId}
Service: reconciliation
Who/scope: recon:read
Description: One run with its discrepancy items.
Request body:
(none)
Response body:
{
  "runId": "run_01",
  "status": "COMPLETED",
  "items": [
    {
```

```
      "id": "exc_01",
      "type": "OVERPAYMENT",
      "status": "OPEN"
    }
  ]
}
```

```
GET /api/v1/exceptions/{id}
Service: reconciliation
Who/scope: exception:read
Description: One discrepancy with linked property, verification, bill and payment.
Request body:
(none)
Response body:
{
  "id": "exc_01",
  "type": "OVERPAYMENT",
  "status": "OPEN",
  "billNumber": "BILL-2026-GA-0891",
  "paymentId": "pay_03"
}
```

```
POST /api/v1/exceptions/{id}/resolve
Service: reconciliation
Who/scope: exception:resolve (oversight user)
Description: Close a discrepancy. A named user signs off.
Request body:
{
  "resolution": "REFUNDED",
  "resolutionRef": "REFUND-8841"
}
Response body:
{
  "id": "exc_01",
  "status": "RESOLVED",
  "resolvedBy": "USR-0007"
}
Notes: Resolutions: REFUNDED, CREDIT_CARRIED_FORWARD, VERIFICATION_ACCEPTED,
ADJUSTMENT. There is no write-off option.
```

```
POST /api/v1/exceptions/{id}/dismiss
Service: reconciliation
Who/scope: exception:resolve
```

Description: Dismiss with a reason.
Request body:

```
{
  "reason": "Duplicate of exc_00"
}
```

Response body:

```
{
  "id": "exc_01",
  "status": "DISMISSED"
}
```

GET /api/v1/dashboards/revenue?countryCode=GH&assemblyCode=&period=2026
Service: reporting
Who/scope: dashboard:read (oversight)
Description: Assessments, confirmed collections, outstanding balances and collection efficiency. Oversight-only, read-only.
Request body:
(none)
Response body:

```
{
  "period": "2026",
  "assessed": {
    "count": 4200,
    "amountDue": "5040000.00"
  },
  "collected": {
    "amount": "3100000.00"
  },
  "outstanding": "1940000.00",
  "collectionEfficiency": "61.5%",
  "byAssembly": [
    {
      "assemblyCode": "GEMA",
      "collected": "412000.00",
      "outstanding": "88000.00"
    }
  ]
}
```

Notes: Numerator, denominator and period are fixed per KPI so figures are comparable across assemblies.

GET /api/v1/dashboards/properties?countryCode=GH&assemblyCode=GEMA&paymentStatus=UNPAID
Service: reporting

Who/scope: dashboard:read (oversight)
Description: Properties by assessed status and bill payment status, for oversight tallying.
Request body:
(none)
Response body:

```
{
  "total": 4200,
  "byPropertyStatus": {
    "ASSESSED": 600,
    "UNASSESSED": 3600
  },
  "byPaymentStatus": {
    "PAID": 1800,
    "PARTIALLY_PAID": 900,
    "UNPAID": 1500
  }
}
```

GET /api/v1/audit-log?actorCode=USR-0007&action=VIEW&from=2026-10-01
Service: operations
Who/scope: audit:read (oversight admin)
Description: Every oversight login and every screen or record an oversight user viewed, with actor, time and what was looked at.
Request body:
(none)
Response body:

```
{
  "items": [
    {
      "id": "aud_01",
      "actorCode": "USR-0007",
      "action": "VIEW",
      "resource": "dashboards/revenue",
      "at": "2026-10-15T09:00:00Z"
    },
    {
      "id": "aud_02",
      "actorCode": "USR-0007",
      "action": "LOGIN",
      "at": "2026-10-15T08:58:00Z"
    }
  ],
  "page": 1,
  "nextPage": null,
```

```
  "total": 2
}
Notes: Written automatically by the auth hook on every oversight request;
oversight users cannot opt out or self-exempt.

---------------------------------------------------------------------
GET /api/v1/admin/outbox?status=failed
Service: operations
Who/scope: ops:admin (core admin)
Description: Failed or stuck events in a service outbox, for support.
Request body:
(none)
Response body:
{
  "items": [
    {
      "id": "evt_01",
      "eventType": "property.verified.v1",
      "attempts": 8,
      "lastError": "broker unavailable"
    }
  ],
  "page": 1,
  "nextPage": null,
  "total": 1
}
Notes: Each service exposes this for its own outbox.

---------------------------------------------------------------------
POST /api/v1/admin/outbox/{id}/replay
Service: operations
Who/scope: ops:admin
Description: Replay an event after approval.
Request body:
{
  "approvalRef": "CHG-2026-114"
}
Response body:
{
  "id": "evt_01",
  "status": "QUEUED"
}
Notes: Consumers ignore duplicates through their inbox.

---------------------------------------------------------------------
GET /health/live, /health/ready, /metrics, /docs
```

Service: operations

Who/scope: none (internal network)

Description: Every service exposes liveness, readiness, Prometheus metrics and its OpenAPI page.

Request body:

(none)

Response body:

```
{
  "status": "ok"
}
```